THE EVOLUTION OF LEISURE THROUGH TECHNOLOGY

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https://github.com/olehhrozyk22/time-use-analysis-project

MOTIVATION

Technology Changing Behavior

- Technology has been integrated into daily routines and leisure activities
- Over the past two decades, there has been a rapid growth of smartphones and social media use

Influence on Social Interaction

- Screen use may alter how people interact and communicate with others
- Question whether digital engagement complements or replaces in-person interaction

Concern Over Future

- Increased access to digital entertainment may shift leisure preferences
- Understanding these trends helps evaluate how technology may shape future behavior

DATA SOURCE AND STRUCTURE

Dataset: American Time Use Survey (ATUS)

- Conducted by the U.S. Bureau of Labor Statistics
- Measures how Americans allocate their time during a 24-hour day
- Respondents report detailed activity episodes and durations
- Link: https://www.atusdata.org/atus/

Overview

- Analyzed the years 2003 to 2024
- Thousands of activity records per year, where each row represents an activity episode

Key Variables

- YEAR – survey year
- ACTIVITY – activity code
- DURATION – minutes spent in the activity

Activities Categorized Into:

- Screen activities
- Social interaction
- Non-screen leisure
- Sports (playing vs watching)

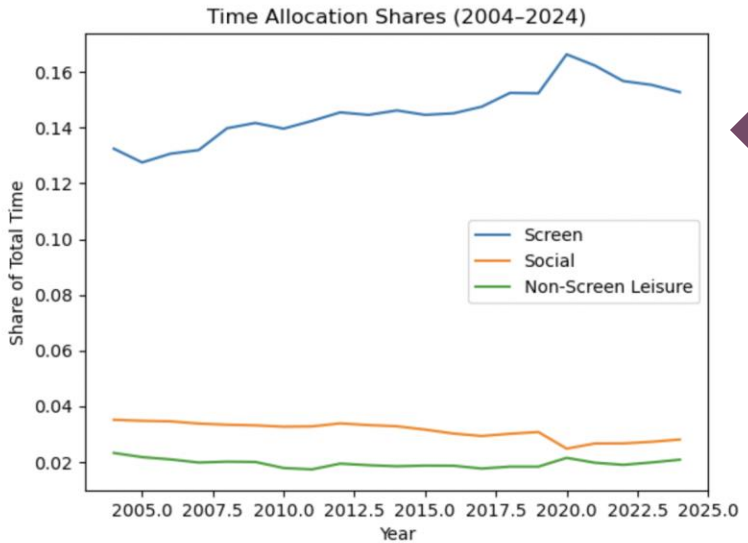
RESEARCH QUESTION ONE

How has increased access to digital technology altered the allocation of time between screen-based activities, traditional leisure activities, and in-person social interaction over the last 20 years?

Null Hypothesis (H0):
Technology has no effect on the relative time spent in screen, social, and non-screen leisure activities.

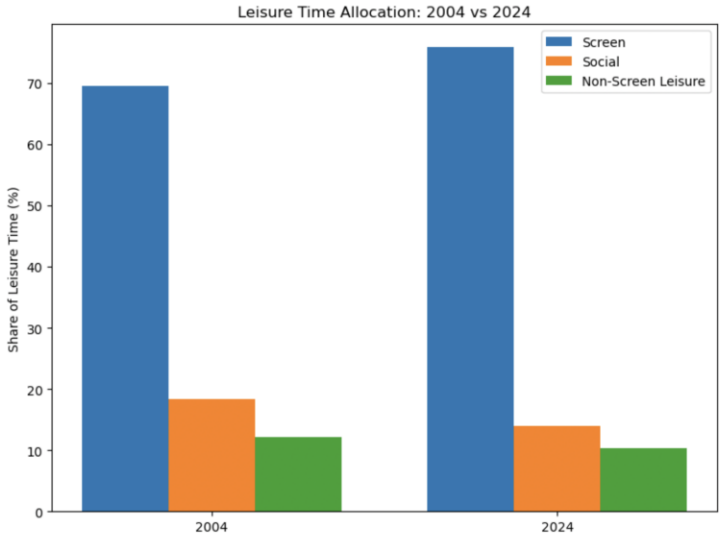
Alternative Hypothesis (H1):
Technology has affected the allocation of time between screen, social, and non-screen leisure activities.

CHANGES IN LEISURE ACTIVITY OVER TIME



Line Chart

Bar Chart



Rise in Screen Time

- Screen activity shows a clear upward trend over the sample period
- Social and non-screen leisure activities show slower growth or relatively stable patterns

Shift in Leisure Time Allocation

- Screen-based leisure increased from 70% to 75% of total leisure time, indicating growing influence of technology
- Social and non-screen leisure declined modestly, suggesting a shift toward digital forms of leisure

STATISTICAL EVIDENCE: LEISURE ACTIVITIES

Pivot Table

activity_type year	non_screen_leisure	other	screen	social
2003	64.626664	69.394535	98.420788	79.153633
2004	64.733777	69.266843	96.992867	76.114175
2005	63.916549	68.822346	94.795581	73.666855
2006	62.364190	68.262323	93.818763	72.652228
2007	64.162341	71.246139	98.325659	77.555628
2008	66.230686	69.094761	98.433321	75.342543
2009	65.254621	69.305789	97.991004	74.089812
2010	68.310879	70.889030	101.867484	79.020041
2011	70.604630	73.060015	105.332647	83.602387
2012	72.712759	73.793285	106.815634	84.490798
2013	71.447612	72.359652	105.097132	81.469321
2014	67.910998	71.219424	103.503520	77.850171
2015	67.785516	69.902482	100.402069	74.947122
2016	68.477361	69.494913	100.570957	75.472935
2017	67.983469	69.938560	104.242525	75.840957
2018	71.344059	70.928217	107.163809	78.914942
2019	71.171920	70.111817	106.866643	77.899795
2020	78.649550	77.154567	111.854044	82.693791
2021	74.397517	75.097727	110.535911	86.858533
2022	74.402750	75.707707	111.796433	88.076771
2023	77.459777	75.955593	112.989780	88.403907
2024	81.088693	74.713835	110.913083	88.951710

Correlation

activity_type activity_type	screen	social	non_screen_leisure
screen	1.000000	0.874685	0.945313
social	0.874685	1.000000	0.865436
non_screen_leisure	0.945313	0.865436	1.000000

Average Time Spent in Leisure Activities (2003–2024):

- Screen activities consistently have the highest average minutes per episode, increasing from about 97 minutes to over 110 minutes over the sample period
- Social and non-screen leisure activities show moderate growth but remain below screen activity durations
- Strong positive correlations between activities
- Indicates leisure activities tend to move together rather than screen time replacing other activities

Regression

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 social   R-squared:                       0.765
Model:                            OLS   Adj. R-squared:                  0.753
Method:                 Least Squares   F-statistic:                     65.13
Date:                Thu, 12 Mar 2026   Prob (F-statistic):           1.02e-07
Time:                        23:58:06   Log-Likelihood:                -50.699
No. Observations:                  22   AIC:                             105.4
Df Residuals:                      20   BIC:                             107.6
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.2770      9.854      0.028      0.978     -20.279      20.833
screen         0.7666      0.095      8.070      0.000       0.568       0.965
==============================================================================
Omnibus:                        2.705   Durbin-Watson:                   0.536
Prob(Omnibus):                  0.259   Jarque-Bera (JB):                1.586
Skew:                          -0.390   Prob(JB):                        0.453
Kurtosis:                       1.940   Cond. No.                     1.89e+03
==============================================================================
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- Screen time is a statistically significant predictor of social activity ($R^2 = 0.765$), and since $p < 0.05$, we reject the null hypothesis that technology has no effect on social activity

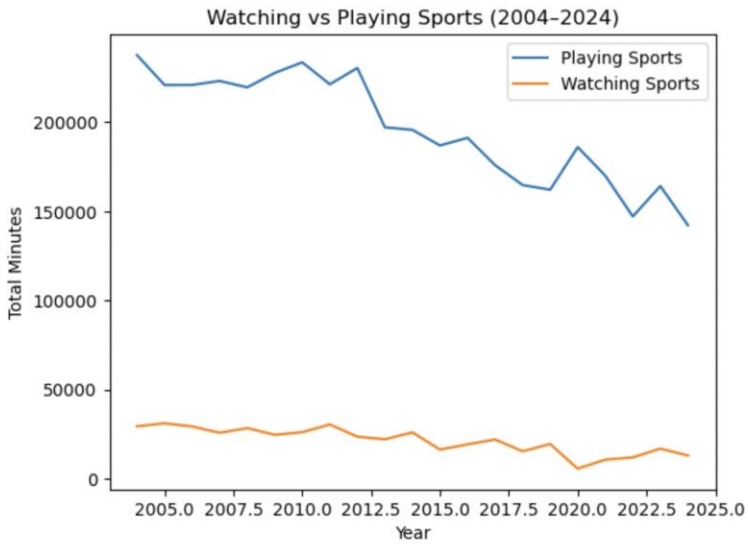
RESEARCH QUESTION TWO

How has increased access to streaming services affected how much time individuals spend playing sports vs watching them over the last 20 years?

Null Hypothesis (H0):
Technology has no effect on the relationship between playing sports and watching sports.

Alternative Hypothesis (H1):
Technology has changed the relationship between playing sports and watching sports.

SPORTS PARTICIPATION VS VIEWERSHIP



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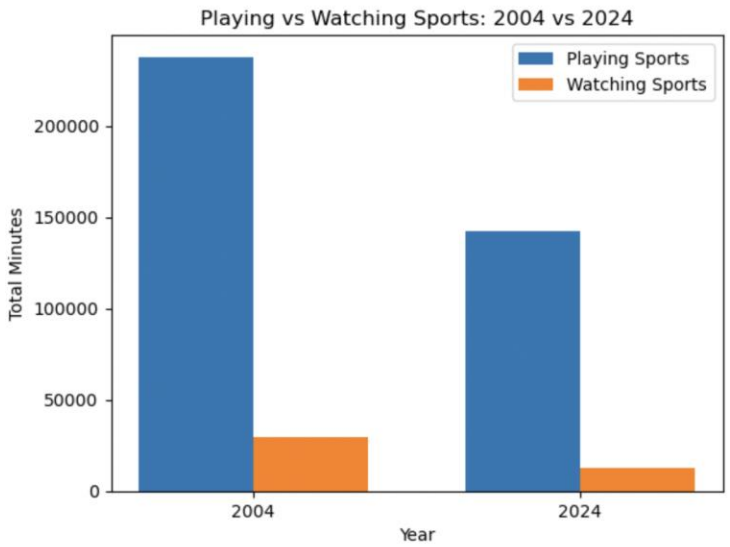

Significant Decline in Playing Sports

- Playing sports shows a gradual decline over the sample period, while watching sports remains relatively stable with moderate variation
- Technology may make sports content more accessible through streaming, but the data does not show a strong shift from participation to viewing

Changes in Participation and Viewership

- Playing sports declined substantially over the period, suggesting reduced participation in physical activities
- Watching sports also decreased, but remained far lower than participation levels, indicating no clear shift from playing to watching sports

STATISTICAL EVIDENCE: SPORTS ENGAGEMENT

Pivot Table

sports_type	playing_sports	watching_sports
year		
2003	78.500218	147.322493
2004	77.127839	150.200000
2005	75.245232	146.009434
2006	77.660457	155.271277
2007	79.016637	162.354430
2008	75.313100	167.284024
2009	76.258540	149.335366
2010	79.425561	139.596774
2011	80.116178	157.838542
2012	81.758425	155.231788
2013	76.809895	144.559211
2014	71.385850	163.708861
2015	66.563190	142.052632
2016	70.400810	143.939850
2017	66.991241	145.860000
2018	67.518852	143.745283
2019	66.698067	157.186992
2020	70.746768	124.659091
2021	64.670217	146.722222
2022	68.137564	173.470588
2023	65.759311	184.318681
2024	65.014175	151.070588

Average Time Spent Playing and Watching Sports (2003-2024):

- Correlation between playing and watching sports is r = 0.899
- Strong positive relationship suggests sports engagement moves together over time
- Years with more participation also tend to have more viewership
- Evidence does not support a simple substitution from playing to watching sports
- Results suggest broader trends in overall sports interest rather than technological replacement

Regression

```
                            OLS Regression Results
==============================================================================
Dep. Variable:        watching_sports   R-squared:                       0.809
Model:                            OLS   Adj. R-squared:                  0.799
Method:                 Least Squares   F-statistic:                     84.59
Date:                Sun, 15 Mar 2026   Prob (F-statistic):           1.27e-08
Time:                        15:19:30   Log-Likelihood:                -215.12
No. Observations:                  22   AIC:                             434.2
Df Residuals:                      20   BIC:                             436.4
Df Model:                           1
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const          -1.739e+04   4459.956     -3.898      0.001   -2.67e+04   -8082.543
playing_sports     0.1968      0.021      9.197      0.000       0.152       0.241
==============================================================================
Omnibus:                       13.477   Durbin-Watson:                   2.092
Prob(Omnibus):                  0.001   Jarque-Bera (JB):               12.570
Skew:                          -1.388   Prob(JB):                      0.00186
Kurtosis:                       5.452   Cond. No.                     9.74e+05
==============================================================================
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- Playing sports is a statistically significant predictor of watching sports ($R^2 = 0.809$) and since $p < 0.001$, we reject the null hypothesis that technology has no effect on the relationship between sports participation and viewership

CONCLUSION

- Screen-based leisure activities have increased over time, indicating the growing role of technology in daily leisure behavior
- Regression results show a strong relationship between screen and social activity, suggesting technology may complement rather than replace social interaction
- Sports participation has declined over time, while sports viewership has fluctuated but remains strongly related to participation
- Overall, technology appears to reshape how people engage with activities rather than completely replacing traditional behavior
- Sports organizations may benefit from integrating digital content with participation opportunities to maintain interest in both playing and watching sports